TOPSTECHNOLOGIES
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Network Device

1. **Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.**

ANS :

```
                Internet
                   |
                [Router]
                   |
                [Switch]
              /    |    \
            /      |      \
     [Laptop] [Smart TV] [Access Point]
                    |
              [Smartphone]
```

Connections:

- Router → Connects home network to the internet.
- Switch → Connects multiple devices in the local network.
- Laptop → Connected to switch with Ethernet cable.
- Smart TV → Connected to switch with Ethernet cable.
- Smartphone → Connected wirelessly through the Access Point (WiFi).

2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

ANS :

Device	**Purpose**	**Real-World Use Case**
Router	Connects network to the internet	Using WiFi to watch YouTube videos
Switch	Connects multiple devices in a LAN	Connecting PCs for LAN gaming
Hub	Broadcasts data to all devices	Small test network in a lab
Access Point	Provides wireless network access	Connecting smartphones and tablets to WiFi

3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

ANS :

When I order food on Zomato using my phone:

1. My phone sends the request through WiFi to the Access Point.
2. The Access Point forwards the data to the Switch.
3. The Switch sends the data to the Router.
4. The Router forwards the request to the internet.
5. The request reaches Zomato's servers.
6. The response follows the same path back to my phone.

Roles:

- Access Point: Provides WiFi connection.
- Switch: Transfers data between local devices.
- Router: Connects the home network to the internet.

4. Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Hint : Think about devices like switches, powerline adapters, or WiFi extenders.

ANS :

Option 1: Powerline Adapter

- One adapter plugs into a wall socket near the router.
- Another adapter plugs into a wall socket near the PC.
- Data travels through the home's electrical wiring.
- The PC connects using an Ethernet cable.

Option 2: WiFi Extender

- Place the WiFi extender between the router and PC.
- The extender receives and boosts the WiFi signal.
- Connect the PC to the extender using an Ethernet cable (if supported).

Result:

Both options allow the gaming PC to access the internet even when it is far from the router.